Mapping Off-Targets in Transcriptional Regulation Experiments using Callpeak Analysis

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Introduction

CRISPR/Cas9 and related technologies have been used to repress and activate individual transcripts by using single guide RNA (sgRNA) to localize effector domains to the transcription start site of a target sequence. By fusing a transcriptional repression domain (KRAB) to catalytically inactive Cas9 protein (dCas), CRISPRi can selectively repress gene function (Gilbert et al., 2014). We observed that when conducting genetic screens using CRISPRi, individual sgRNAs reproducibly produced strong knockdown phenotypes that were not present in other sgRNAs targeting the same gene. These observations raise an important question: are off-target interactions between dCas9-KRAB-sgRNA complexes interacting with known or novel genes in a pathway? Through ChIP-seq we mapped the binding of several dCas-9 KRAB-sgRNA complexes genome wide. This allowed us to observe binding events at both on- and off-target sites. We developed MOTREC to highlight off-target binding sites that are false positives and those that are potentially novel.

Results

Using ChIP-seq, we observed that phenotypes can be different for distinct sgRNAs targeting the same gene in CRISPRi Knockdown experiments (Figure 1A). We observed that SOCS3 guides show high variability between replicates along with equivalent recruitment of Cas9-KRAB to the 5′ UTR of the target gene (Figure 1B). Secondary analysis using a peak calling algorithm quantifies binding events of dCas9-Krab with the genome. At a fold enrichment minimum of 5, we observed 27 off-targets for guide 4 (SOCS3_-_76355958.23-P1P2) and 281 off-targets for guide 1 (SOCS3_-_76356125.23-P1P2) (Figure 1C). These off-targets can then be mapped back to the genome using MOTREC which provides a list of each off-target binding event and the closest feature (Figure 1D). The closest feature is calculated by finding the absolute value of the difference between peak midpoint and feature start. Gene ontology can be used to flag features classified under unexpected GO identifiers. Using MOTREC on SOCS3 guides revealed binding events between SOCS3g1 and PEX19 (Figure 1E).

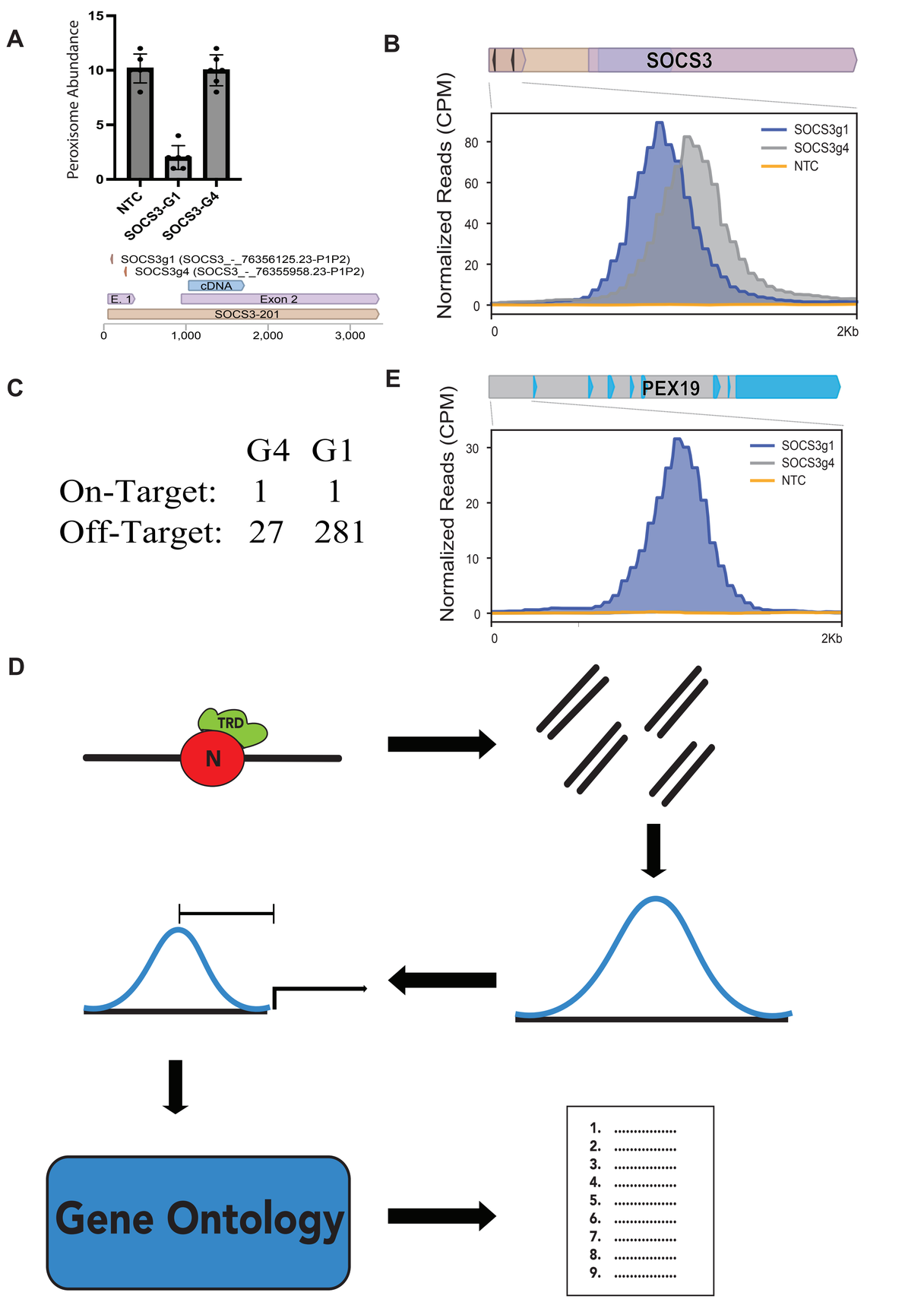

Figure 1: ChIP-seq of dCas9-KRAB can diagnose artifacts in transcriptional regulation experiments. (A) CRISPRi knockdown cell lines show significant replicate-replicate phenotypic variability. (B) CRISPRi knockdown cell lines show equivalent recruitment of dCas9-KRAB to 5'UTR of target gene. (C) Individual CRISPRi guides bind off-target sites in the genome. (D) MOTREC workflow. (E) A CRISPRi guide targeting SOCS3 binds to PEX19.

We also demonstrated that dCas9-KRAB binding can be used to measure off-target sites for Cas9 cutting guides. This was done using VEGFA site 2, a guide known to bind many genomic off-target sites. We compared our approach with other off-target detection methods Discover-seq and Discover-seq+ (Weinert et al., 2019, Zou et al., 2023). These methods track the presence of DNA repair factor MRE11 to detect the locations where Cas9 is creating double strand breaks (DSBs). To compare the three approaches, Discover-seq+ results were annotated at locations of the genome Discover-seq and MOTREC also identified. Results were then ordered by Discover score. MOTREC identified both on- and off-target sites where Cas9 cutting guides were active. Four of the top five off-target sites were identified using MOTREC based on Discover score (Figure 2).

Location	D+	D	M	Score
chr6:43768825-43772825	✓	✓	✓	417
chr5:6713005-6717005	✓	✓	✓	191
chr2:241273191-241277191	✓	✓	✓	157
chr11:31793935-31797935	✓	✓	✓	129
chr1:151057409-151061409	✓	✓		104
chr4:38534006-38538006	✓	✓	✓	100
chr5:139646671-139650671	✓	✓		92
chr18:23777595-23781595	✓	✓	✓	88
chr10:133334443-133338443	✓	✓	✓	84
chr17:41886504-41890504	✓	✓		50
chr9:100835367-100839367	✓	✓	✓	44
chr2:12602649-12606649	✓	✓		43
chr11:372667-376667	✓	✓		42
chr22:50444373-50448373	✓	✓		30
chr9:136544395-136548395	✓	✓		25
Total	187	87	14	

Figure 2: dCas9-KRAB binding can be used to measure off-target sites for Cas9 cutting guides

Conclusion

Pooled screen data processing pipelines currently censor data in which multiple guides targeting the same gene have different phenotypes. This can hide potentially interesting off-target interactions or prevent identification of poorly annotated reference genomes. MOTREC recovers this censored data by categorizing sgRNAs based on the location of the off-target interactions.

Genome-wide transcriptional regulation has become an essential genetic tool. Our approach lays the groundwork for systematic interrogation and improvement of these technologies. For example, MOTREC of all 100,000 guides in the genome-wide library using single cell approaches would dramatically improve the quality and annotation of knockdown approaches.

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